

Data Usage Guide — EMS Readiness Optimization

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Purpose: Tell you exactly which files are current, which are legacy, and which to use for any given analysis task.

Quick Rules

Rule	Details
Source of truth for simulation	results/baseline/ — cap=2, spatially-stratified P0, 30 replications
Source of truth for optimization	results/baseline/ — cap=2 (current default per DEC-010). For historical cap=5 results, see results/archive/optimization/ (Phase 3 legacy)
Source of truth for publication tables	results/tables/table1_* through table4_*
Default capacity assumption	cap=2 (see DEC-010). Files with cap=5 are either legacy or part of capacity sweeps
P0 means	Spatially-stratified uniform (latitude-based). Files with P0 without _spatial suffix in capacity_comparison/ or heatmaps/ refer to the deprecated index-based P0
Legacy indicator	Any file with _legacy in the name, or P0 without _spatial in capacity/heatmap sweeps

Folder-by-Folder File Inventory

results/baseline/ —  **PRIMARY SOURCE OF TRUTH**

This is the canonical simulation output. All files use **cap=2**, **spatially-stratified P0**, and **30 Monte Carlo replications**.

Path	Status	Description
allocations/allocations_K{10..48}.csv	✓ CURRENT	Unit allocations for P0/P1/P2 at each K
simulation/results_K{10..48}.csv	✓ CURRENT	Raw simulation output per K
simulation/all_results_raw.csv	✓ CURRENT	Combined raw results across all K
tables/descriptive_statistics.csv	✓ CURRENT	Mean RT, CI, coverage, utilization per (K, policy)
tables/anova_results.csv	✓ CURRENT	ANOVA test results
tables/posthoc_comparisons.csv	✓ CURRENT	Tukey HSD pairwise comparisons
tables/effect_sizes.csv	✓ CURRENT	Cohen's d effect sizes
tables/confidence_intervals.csv	✓ CURRENT	95% confidence intervals
tables/queue_statistics.csv	✓ CURRENT	Queue/utilization metrics
figures/*.png	✓ CURRENT	All production figures (allocation maps, RT distributions, sensitivity curves)
comparison_with_v1.csv	📊 COMPARISON	Shows how v2 (cap=2) compares to v1 (cap=5)
experiment_log.txt	✓ CURRENT	Run metadata and parameters

results/archive/optimization/ — ⚠️ PHASE 3 HISTORICAL (cap=5)

These are the original Phase 3 optimization-only results. They use **cap=5** (the old default before DEC-010). The optimization metrics (objective values, coverage) are mathematically correct for cap=5 but **do not reflect the current cap=2 default**. Use `results/baseline/` for cap=2 results.

Path	Status	Description
<code>allocations_K{15..48}.csv</code>	⚠ LEGACY	Allocations at cap=5
<code>policy_comparison.csv</code>	⚠ LEGACY	Full metrics at cap=5 — correct for cap=5, not current default
<code>sensitivity_analysis.csv</code>	⚠ LEGACY	RT & coverage by K at cap=5
<code>findings_summary.json</code>	⚠ LEGACY	Best-policy summary at cap=5
<code>PHASE3_SUMMARY.md</code> / <code>.pdf</code>	⚠ LEGACY	Phase 3 write-up (cap=5 era). Conclusions still directionally valid

Why keep these? They document the Phase 3 milestone and remain useful for comparing cap=5 vs cap=2 outcomes.

results/tables/ — Mixed Current & Legacy

Path	Status	Description
table1_baseline_comparison.csv / .tex	✓ CURRENT	Publication Table 1 — P0 vs P1 vs P2 at K=20,30,40
table2_anova_summary.csv / .tex	✓ CURRENT	Publication Table 2 — ANOVA results
table3_pairwise_comparisons.csv / .tex	✓ CURRENT	Publication Table 3 — Tukey HSD
table4_sensitivity_summary.csv / .tex	✓ CURRENT	Publication Table 4 — Fleet/demand sensitivity
exp1_summary.csv	✓ CURRENT	Experiment 1 policy comparison summary
exp2_pivot_rt.csv	✓ CURRENT	Experiment 2 fleet sensitivity pivot (current P0 only)
exp3_pivot_rt.csv	✓ CURRENT	Experiment 3 demand sensitivity pivot
exp4_pivot_rt.csv	✓ CURRENT	Experiment 4 service robustness pivot
exp2_pivot_rt_with_legacy.csv	📊 COMPARISON	Fleet sensitivity with both current P0 and legacy P0 columns
production_results.csv	✓ CURRENT	Combined production simulation results
optimization_comparison.csv	⚠️ LEGACY	Optimization-only comparison (cap=5 era)
optimization_results.csv	⚠️ LEGACY	Optimization-only results (cap=5 era)
sensitivity_summary.csv	✓ CURRENT	Sensitivity analysis summary
statistical_analysis.csv	✓ CURRENT	Statistical test results
cbd_comparison.csv	✓ CURRENT	CBD vs non-CBD comparison
cbd_robustness.csv	✓ CURRENT	CBD robustness analysis
cbd_summary_all.csv	✓ CURRENT	CBD comprehensive summary

Path	Status	Description
queue_metrics.csv	✓ CURRENT	Queue performance metrics
queue_anova.csv	✓ CURRENT	Queue ANOVA analysis
queue_statistics.csv	✓ CURRENT	Queue descriptive stats
seasonal_analysis.csv	✓ CURRENT	Seasonal demand patterns
validation_results.csv	✓ CURRENT	V&V validation results
anova_results.csv	✓ CURRENT	ANOVA (production results, cap=2)
confidence_intervals.csv	✓ CURRENT	CI's (from production run)
confid- ence_intervals_notebook.csv	⚠ LEGACY	CI's from older notebook run
descriptive_statistics.csv	✓ CURRENT	Descriptive stats
effect_sizes.csv	✓ CURRENT	Effect sizes
posthoc_comparisons.csv	✓ CURRENT	Posthoc tests
pos- thoc_comparisons_notebook.c sv	⚠ LEGACY	Posthoc from older notebook run

results/figures/ — ✓ Mostly Current

All figures were regenerated or verified during the Phase 10 audit. Publication-quality figures use the `pub_fig*` prefix.

Path Pattern	Status	Description
pub_fig1_policy_comparison.png	✓ CURRENT	Publication Figure 1
pub_fig2_fleet_sensitivity.png	✓ CURRENT	Publication Figure 2
pub_fig3_demand_robustness.png	✓ CURRENT	Publication Figure 3
pub_fig4_service_sensitivity.png	✓ CURRENT	Publication Figure 4
pub_fig5_performance_heatmap.png	✓ CURRENT	Publication Figure 5
fig_* prefix files	✓ CURRENT	Technical report figures (crash heatmap, demand, fire-houses, etc.)
exp1_* through exp4_*	✓ CURRENT	Experiment result plots
validation_* / verification_*	✓ CURRENT	V&V output plots
capacity_sensitivity_heatmap.png	⚠ KNOWN ISSUE	Shows “Data format issue” — placeholder, not usable
capacity_sensitivity_heatmap_notebook.png	✓ CURRENT	Working version from notebook
fleet_sensitivity_v2_dual.png	✓ CURRENT	Fleet sensitivity (v2, cap=2)
fleet_sensitivity_dual.png	⚠ LEGACY	Older fleet sensitivity plot
fleet_sensitivity_curve.png	⚠ LEGACY	Older single-axis fleet sensitivity
policy_comparison.png	✓ CURRENT	General policy comparison
policy_comparison_panel_K20_cap2.png	✓ CURRENT	Panel figure at K=20, cap=2
optimization/*.png	⚠ LEGACY	Phase 3 optimization visualizations (cap=5)

Path Pattern	Status	Description
cbd_*	✓ CURRENT	CBD analysis figures
queue_*	✓ CURRENT	Queue analysis figures
seasonal_*	✓ CURRENT	Seasonal pattern figures
p0_spatial_*	✓ CURRENT	Spatial P0 analysis
production_fleet_sensitivity.png	✓ CURRENT	Production fleet sensitivity
project_summary_dashboard.png	✓ CURRENT	Overview dashboard

results/analysis/capacity_comparison/ — ✓ CURRENT (Capacity Sweep)

Full capacity sensitivity sweep across $\text{cap}=\{1,2,3,4,5\} \times K=\{20,30,40\} \times \{P0, P0_spatial, P1, P2\}$. All data is current and correct.

Path Pattern	Status	Description
alloca- tion_P0_spatial_K*_cap*.csv	✓ CURRENT	Spatially-stratified P0 allocations
allocation_P0_K*_cap*.csv	📊 COMPARISON	Index-based (legacy) P0 allocations — included for comparison only
alloca- tion_P1_demand_K*_cap*.csv	✓ CURRENT	P1 allocations across capacities
alloca- tion_P2_optimised_K*_cap*.csv	✓ CURRENT	P2 optimized allocations
simulation_results.csv	✓ CURRENT	Full simulation results for K=20 sweep
simulation_results_K30.csv	✓ CURRENT	Full simulation results for K=30 sweep
full_comparison.csv	✓ CURRENT	Combined comparison table
optimal_configurations.csv	✓ CURRENT	Best configurations identified
*.png figures	✓ CURRENT	Capacity comparison visualizations

Note: Files with `P0` (no `_spatial`) are the deprecated index-based P0 included for comparison purposes. Use `P0_spatial` files for the canonical baseline.

results/analysis/heatmaps/ — ✓ CURRENT (Comprehensive Sweep)

Heatmap visualizations across $K=\{5,10,15,20,25,30,35,40,45\} \times \text{cap}=\{1,2,3,5\} \times \{P0, P0_spatial, P1, P2\}$. All generated from a single consistent run.

Path Pattern	Status	Description
heat-map_K*_policyP0_spatial_cap*.png	✓ CURRENT	Canonical P0 heatmaps
heat-map_K*_policyP0_cap*.png	📊 COMPARISON	Legacy P0 heatmaps (index-based, for comparison)
heat-map_K*_policyP1_cap*.png	✓ CURRENT	P1 heatmaps
heat-map_K*_policyP2_cap*.png	✓ CURRENT	P2 heatmaps
allocations/*.csv	✓ CURRENT	Underlying allocation data for each heatmap
generation_summary.json	✓ CURRENT	Metadata about the heatmap generation run

Same note: `policyP0` (without `_spatial`) = deprecated index-based P0. Use `policyP0_spatial` for the canonical baseline.

results/analysis/cbd_focused_comparison/ — ✓ CURRENT

CBD-focused optimization experiment results. Demonstrates that CBD weighting does not improve CBD RT but degrades non-CBD RT.

Path	Status	Description
allocations.csv	✓ CURRENT	CBD-focused allocations
comparison_table.csv	✓ CURRENT	CBD vs standard comparison
*.png figures	✓ CURRENT	CBD comparison visualizations
experiment_log.txt	✓ CURRENT	Experiment metadata

results/analysis/distance_comparison/ — ✓ CURRENT

Manhattan vs Haversine distance metric comparison. Shows P2 is robust to distance metric choice.

Path	Status	Description
allocation_comparison.csv	✓ CURRENT	Haversine vs Manhattan allocations
comparison_table.csv	✓ CURRENT	Side-by-side metrics
*.png figures	✓ CURRENT	Distance comparison plots
experiment_log.txt	✓ CURRENT	Experiment metadata

results/simulation/ — ✓ CURRENT

Simulation verification, validation pilots, and production experiment results.

Path	Status	Description
verification/ 01_toy_example.json	✓ CURRENT	Verification test 1
verification/ 02_zero_demand.json	✓ CURRENT	Verification test 2
verification/ 03_single_unit.json	✓ CURRENT	Verification test 3
verification/ 04_extreme_demand.json	✓ CURRENT	Verification test 4
validation_pilot/pi- lot1_*.json/.csv	✓ CURRENT	Pilot 1: P0 vs P2
validation_pilot/pi- lot2_*.json/.csv	✓ CURRENT	Pilot 2: Fleet sensitivity
validation_pilot/pi- lot3_*.json	✓ CURRENT	Pilot 3: Demand sensitivity
production/exp1_*.csv	✓ CURRENT	Experiment 1 results
production/exp2_*.csv	✓ CURRENT	Experiment 2 results
production/exp3_*.csv	✓ CURRENT	Experiment 3 results
production/exp4_*.csv	✓ CURRENT	Experiment 4 results
production/experi- ment_summary.csv	✓ CURRENT	Combined summary
production/experi- ment_log.txt	✓ CURRENT	Run metadata
cbd_experiment/*.csv	✓ CURRENT	CBD experiment results

results/analysis/maps/ — ✓ CURRENT

Static allocation maps at K=40.

Path	Status	Description
map_allocation_P0_K40.png	✓ CURRENT	P0 spatial allocation map
map_allocation_P1_K40.png	✓ CURRENT	P1 allocation map
map_allocation_P2_K40.png	✓ CURRENT	P2 allocation map

Root-Level Files in results/

Path	Status	Description
consistency_verification_report.json	✓ CURRENT	Automated consistency check output
subfolder_audit_report.md / .pdf	✓ CURRENT	Deep audit of all subfolders
file_contents_verification.md / .pdf	✓ CURRENT	File content verification results

Decision Tree: “Which File Should I Use?”

For baseline policy comparison (P0 vs P1 vs P2):

→ results/baseline/tables/descriptive_statistics.csv

→ Or publication-ready: results/baseline/tables/table1_baseline_comparison.csv

For fleet size sensitivity (how does RT change with K?):

→ results/baseline/tables/exp2_pivot_rt.csv (current P0 only)

→ Or with legacy comparison: results/analysis/tables/exp2_pivot_rt_with_legacy.csv

For demand sensitivity:

→ results/baseline/tables/exp3_pivot_rt.csv

For service time robustness:

→ results/baseline/tables/exp4_pivot_rt.csv

For statistical significance:

→ results/baseline/tables/anova_results.csv + posthoc_comparisons.csv + effect_sizes.csv

For capacity sensitivity analysis:

→ results/analysis/capacity_comparison/full_comparison.csv

→ Heatmaps: results/analysis/heatmaps/heatmap_K*_policyP*_spatial_cap*.png

For CBD analysis:

→ `results/analysis/cbd_focused_comparison/comparison_table.csv`

→ Or: `results/analysis/tables/cbd_comparison.csv`

For verification & validation:

→ `results/simulation/verification/*.json`

→ `results/baseline/simulation/validation_pilot/*.json`

For publication figures:

→ `results/figures/pub_fig*.png`

For allocation maps:

→ `results/baseline/figures/allocation_map_K*.png` (cap=2)

→ Or: `results/analysis/maps/map_allocation_P*_K40.png`

Naming Conventions

Pattern	Meaning
<code>_legacy</code> suffix	Explicitly marked as historical/deprecated data
<code>P0</code> in <code>capacity_comparison/</code> or <code>heatmaps/</code> (without <code>_spatial</code>)	Index-based legacy <code>P0</code> — for comparison only
<code>P0_spatial</code> or just <code>P0</code> elsewhere	Current spatially-stratified <code>P0</code>
<code>cap2</code> / <code>cap=2</code>	Current default capacity
<code>cap5</code> / <code>cap=5</code>	Historical default capacity (pre-DEC-010)
<code>_notebook</code> suffix	Generated from older notebook run, not production pipeline
<code>pub_fig*</code>	Publication-quality figures
<code>table{1-4}_*</code>	Publication-ready tables
<code>v2</code>	Current production version (cap=2, spatial <code>P0</code>)

Key Project Decisions Affecting Data

Decision	Impact on Files
DEC-010: Capacity changed from 5 → 2	<code>results/baseline/</code> uses <code>cap=2</code> ; <code>optimization/</code> still shows <code>cap=5</code>
DEC-011: P0 changed to spatially-stratified	All current P0 is spatial; old index-based P0 appears as <code>P0_legacy</code> or <code>P0</code> without <code>_spatial</code> in sweeps
DEC-012: P0 nomenclature standardized	“P0” in any public/current document = spatially-stratified